Manipulating a touch screen with just a single finger is still fairly WIMPy—you just replace the mouse pointer with your stylus or index finger. Multitouch is a bit more complicated. Touching the screen with five fingers is like pushing five mouse pointers across the screen at the same time and clearly changes things for the window manager. Multitouch screens have become ubiquitous and increasingly sensitive and accurate. Nevertheless, it is unclear whether the Five Point Palm Exploding Heart Technique has any effect on the CPU.

## 5.7 THIN CLIENTS

Over the years, the main computing paradigm has oscillated between centralized and decentralized computing. The first computers, such as the ENIAC, were, in fact, personal computers, albeit large ones, because only one person could use one at once. Then came timesharing systems, in which many remote users at simple terminals shared a big central computer. Next came the PC era, in which the users had their own personal computers again.

While the decentralized PC model has advantages, it also has some severe disadvantages. Probably the biggest problem is that each PC has a large SSD or hard disk and complex software that must be maintained. For example, when a new release of the operating system comes out, a great deal of work has to be done to perform the upgrade on each machine separately. At most corporations, the labor costs of doing this kind of software maintenance dwarf the actual hardware and software costs. For home users, the labor is technically free, but few people are capable of doing it correctly and fewer still enjoy doing it. With a centralized system, only one or a few machines have to be updated and those machines have a staff of experts to do the work.

A related issue is that users should make regular backups of their terabyte file systems, but few of them do. When disaster strikes, a great deal of moaning and wringing of hands tends to follow. With a centralized system, backups can be made every night, to disk or even tape (by automated tape robots).

Another advantage is that resource sharing is easier with centralized systems. A system with 256 remote users, each with 16 GB of RAM (or 4 TB in total), will have most of that RAM idle most of the time. With a centralized system with just 1 TB of RAM, it never happens that some user temporarily needs a lot of RAM but cannot get it because it is on someone else's PC. The same argument holds for disk space and other resources.

Finally, we are starting to see a shift from PC-centric computing to Web-centric computing. One area where this shift is very far along is email. People used to get their email delivered to their home machine and read it there. Nowadays, many people log into Gmail, Hotmail, or Yahoo and read their mail there. In addition, many people log into other Websites to do word processing, build spreadsheets, and other things that used to require PC software. It is even possible that eventually the only software most people run on their PC is a Web browser, and maybe not even that.

It is probably a fair conclusion to say that most users want high-performance interactive computing but do not really want to administer a computer. This has led researchers to reexamine timesharing using simple text terminals (now called **thin clients**) that meet modern terminal expectations. X was a step in this direction and dedicated X terminals were popular for a little while but they fell out of favor because they cost as much as PCs, could do less, and still needed some software maintenance. The holy grail would be a high-performance interactive computing system in which the user machines had no software at all. Interestingly enough, this goal is achievable, although existing solutions tend to be less extreme.

One of the best known thin clients is the **ChromeBook**. It is pushed actively by Google, but with a wide variety of manufacturers providing a wide variety of models. The notebook runs **ChromeOS** which is based on Linux and the Chrome Web browser and was originally assumed to be online all the time. Most other software is hosted on the Web in the form of **Web Apps**, making the software stack on the Chromebook itself considerably thinner than in most traditional notebooks. It turns out that model did not work so well, so eventually Google made it possible to run Android apps natively on Chromebooks. On the other hand, a system that runs a full Linux stack, and a Chrome browser, is not exactly anorexic either.

## 5.8 POWER MANAGEMENT

The first general-purpose electronic computer, the ENIAC, had 18,000 vacuum tubes and consumed 140,000 watts of power. As a result, it ran up a nontrivial electricity bill. After the invention of the transistor, power usage dropped dramatically and the computer industry lost interest in power requirements. However, nowadays power management is back in the spotlight for several reasons, and the operating system is playing a role here.

Let us start with desktop PCs. A desktop PC often has a 200-watt power supply (which is typically 85% efficient, that is, loses 15% of the incoming energy to heat). If 100 million of these machines are turned on at once worldwide, together they use 20,000 megawatts of electricity. This is the total output of 20 average-sized nuclear power plants. If power requirements could be cut in half, we could get rid of 10 nuclear power plants. From an environmental point of view, getting rid of 10 nuclear power plants (or an equivalent number of fossil-fuel plants) is a big win for the planet.

The other place where power is a very big issue is on battery-powered computers, including notebooks, smartphones, and tablets, among others. The heart of the problem is that the batteries cannot hold enough charge to last very long, a few hours at most. Furthermore, despite massive research efforts by battery companies, computer companies, and consumer electronics companies, progress is glacial. To